Artificial Intelligence in Radiology

How fast it is growing, where it sits, who the players are, and how much is pediatric

Pediatric Radiology AI project

September 3, 2026

Objectives

- Quantify how much **radiology AI** has grown, from a pre-deep-learning 2008 baseline to today, using reproducible queries against public databases.
- Measure **where** inside radiology the AI work sits — by imaging modality and by the question each model answers.
- Measure how much of radiology AI is **pediatric**, and in which modalities and tasks.
- Identify the **biggest players**: most-cited papers, most-used open-source software, most-cleared commercial products, and who the trade press talks about.
- Summarize what the field **does well**, what is **bleeding edge**, and what remains **unresolved** — for a children's hospital.

Methods — sources, filters, and players

Academic output. PubMed (E-utilities) yearly counts per query and per modality/task term group; **OpenAlex** citation counts (union of modality/task searches, deduped); **DBLP** for ML venues; **PatentsView** for granted patents.

Open-source software. GitHub search leaderboards plus a fixed list of known tools fetched by name (search alone misses MONAI and nnU-Net because their descriptions never say "radiology").

Commercial players. The **FDA AI-enabled device list** (public spreadsheet: decision date, device, company, lead panel), filtered to the Radiology panel: products per year, clearances per company, device names matching pediatric terms; cross-checked against a curated list of products with pediatric indications.

Trade press. Newsletter archives (The Imaging Wire, RSNA News, TLDR, Signify Research, Radiology Business) split into stories; a story is radiology-AI when AI terms co-occur with imaging terms, pediatric when pediatric terms also co-occur; company/tool mentions counted per story (sponsor blocks excluded).

Pediatric filter. PubMed: the radiology-AI query AND (pediatric* OR paediatric* OR child* OR infant* OR neonat* OR adolescen* OR "children's hospital"). Most-cited lists: the title must also carry a pediatric term (bone age, fetal, newborn, . . .), otherwise highly cited adult papers float in. Newsletters: pediatric term inside the same story.

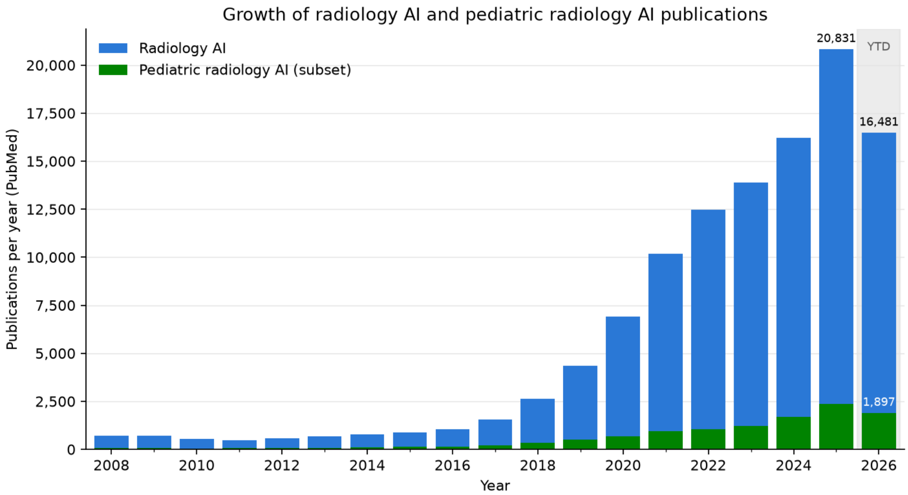
Representative PubMed query (radiology ∩ AI, version 2):

(radiology OR radiograph* OR "medical imaging" OR MRI OR "computed tomography" OR CT[tiab] OR ultrasound[tiab] . . . NOT ("optical coherence" OR fundus OR dental OR histopatholog* . . .)) AND ("artificial intelligence" OR "deep learning" OR "convolutional neural network" OR radiomics . . .)

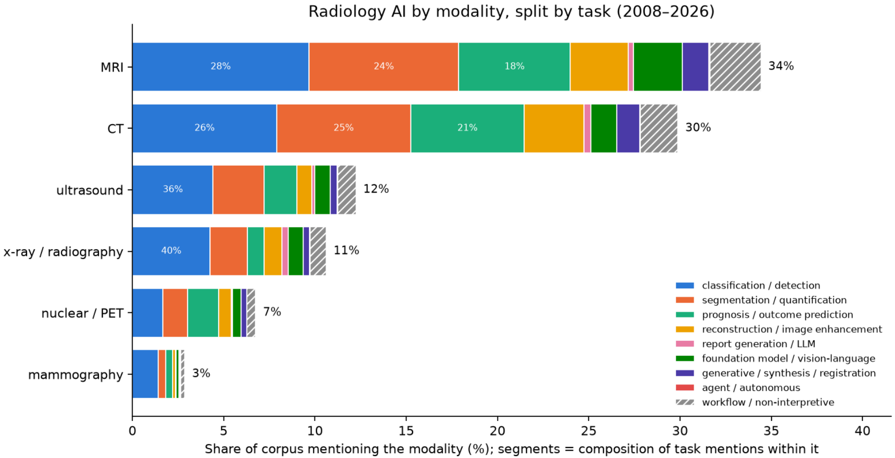
Methods — is the radiology-AI query sufficient?

- **Recall** on 30 hand-picked landmark papers (methods, clinical validations, reconstruction, guidelines): **26/30** retrieved (87%); pediatric subset 11/11. Missed: nnU-Net, Litjens DL survey, MedSAM, Kermany OCT / chest X-ray — methods papers that name no modality ("biomedical image segmentation") or cross-domain papers removed by the non-radiology-imaging exclusion (OCT, microscopy); the most-cited tables use OpenAlex union queries, which recover them.
- **Precision proxy** (2025): a strict title/abstract-only variant of the query returns 18,217 records vs 20,831 for the headline query; 76% of the headline count is confirmed by explicit wording (pediatric: 72%). The remainder enters via MeSH indexing; a sample of those records is in the validation report.
- **Term audit**: PubMed auto-maps bare *ultrasound* to the "diagnostic imaging" MeSH subheading and bare *tomography* to a tree that includes optical coherence tomography. Version 1 of the query counted 63% of radiology-AI papers as "ultrasound"; version 2 fields every modality term ([tiab] or its specific MeSH heading) and excludes ophthalmic, dental, pathology and endoscopic imaging.
- **Verdict**: recall is high; precision was the problem and is now controlled. Recent-year counts still undercount (indexing lag) and the current year is partial.

How much has AI grown in the radiology literature?

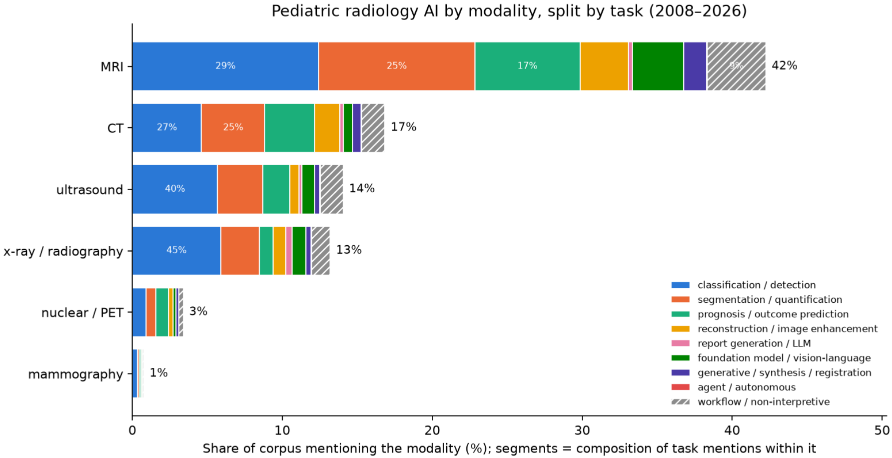


Where the AI work sits — by modality, and which questions are asked



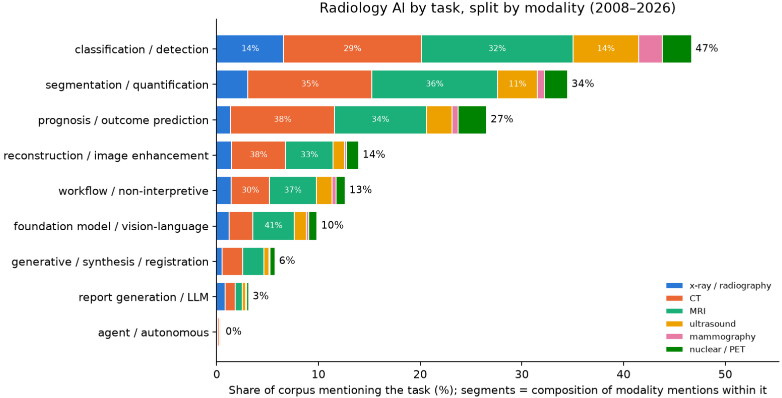
Bar = share of radiology-AI papers whose title/abstract names the modality (2008–present, overlapping). Segments = the mix of task terms inside that modality's papers.

Pediatric radiology AI — by modality, and which questions are asked



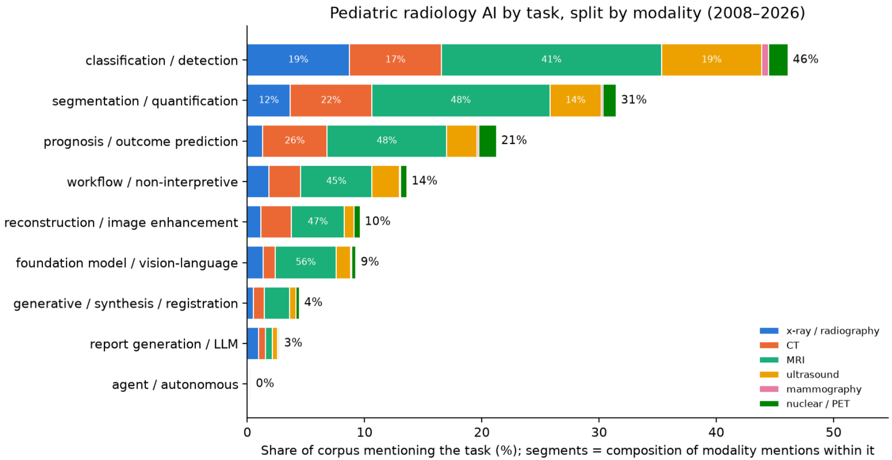
Same construction, restricted to the pediatric subset. Top pediatric modalities: MRI 43%, CT 17%, ultrasound 14%, x-ray / radiography 13% (share of pediatric radiology-AI papers).

Where the AI work sits — by task (what the model produces)



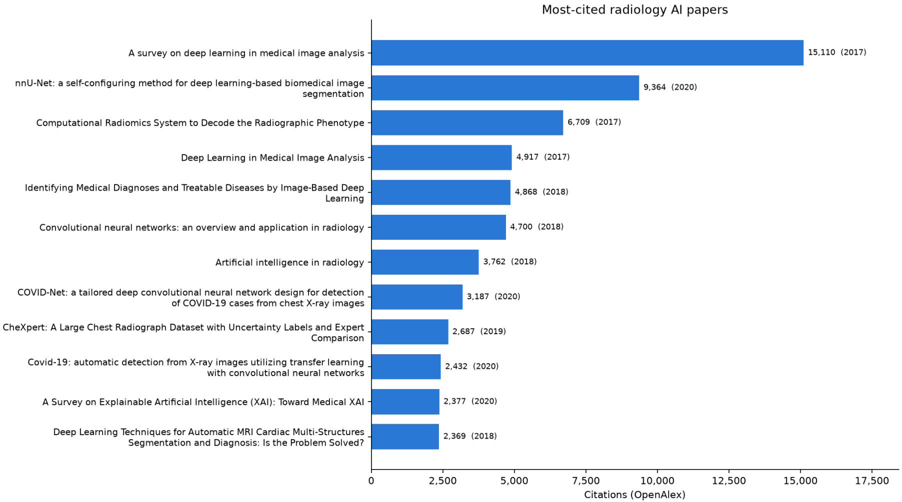
- **classification / detection**: is a finding present, and where (label or box)
- **segmentation / quantification**: outline / measure a structure or lesion
- **prognosis / outcome prediction**: predict risk, response, or survival from images
- **reconstruction / image enhancement**: better images from less dose or shorter scans
- **report generation / LLM**: draft, summarize, or extract from report text
- **foundation model / vision-language**: large pretrained models reused across tasks
- **generative / synthesis / registration**: make or align images (GAN, diffusion)
- **agent / autonomous**: multi-step actions taken without a human in the loop
- **workflow / non-interpretive**: protocoling, scheduling, ordering, education

Pediatric radiology AI — by task



Same task categories, restricted to the pediatric subset. Labels overlap; a paper can be both segmentation and prognosis.

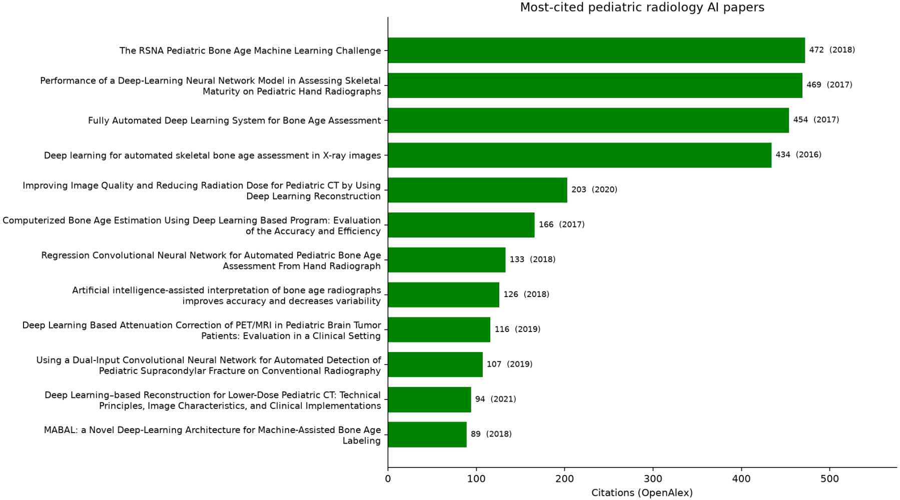
Biggest players — most-cited radiology AI papers



What the most-cited radiology AI papers answer ($\geq$ 1,500 citations)

Cites	Paper	Clinical question it answers
15,110	A survey on deep learning in medical image analysis (2017)	*survey*: What can deep learning do across medical imaging? (the field's reference survey)
9,364	nnU-Net: a self-configuring method for deep learning-based biomedical... (2020)	*segmentation*: Can one self-configuring method segment any organ or lesion without hand tuning? (nnU-Net)
6,709	Computational Radiomics System to Decode the Radiographic Phenotype (2017)	*radiomics*: Can standardized image features (radiomics) be extracted reproducibly? (PyRadiomics)
4,917	Deep Learning in Medical Image Analysis (2017)	*survey*: How does deep learning apply to medical image analysis? (review)
4,868	Identifying Medical Diagnoses and Treatable Diseases by Image-Based D... (2018)	*classification*: Can transfer learning diagnose pediatric pneumonia on chest X-ray and retinal disease on OCT?
4,700	Convolutional neural networks: an overview and application in radiolo... (2018)	*tutorial*: What is a convolutional neural network, explained for radiologists?
3,762	Artificial intelligence in radiology (2018)	*review*: How will AI change radiology practice? (Hosny et al.)
3,187	COVID-Net: a tailored deep convolutional neural network design for de... (2020)	*classification*: Can a CNN detect COVID-19 on chest X-ray? (COVID-Net)
2,687	CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels an... (2019)	*dataset*: A 224k chest-radiograph dataset with uncertainty labels for training and benchmarking (CheXpert)
2,432	Covid-19: automatic detection from X-ray images utilizing transfer le... (2020)	*classification*: Can transfer learning detect COVID-19 on X-ray with small data?

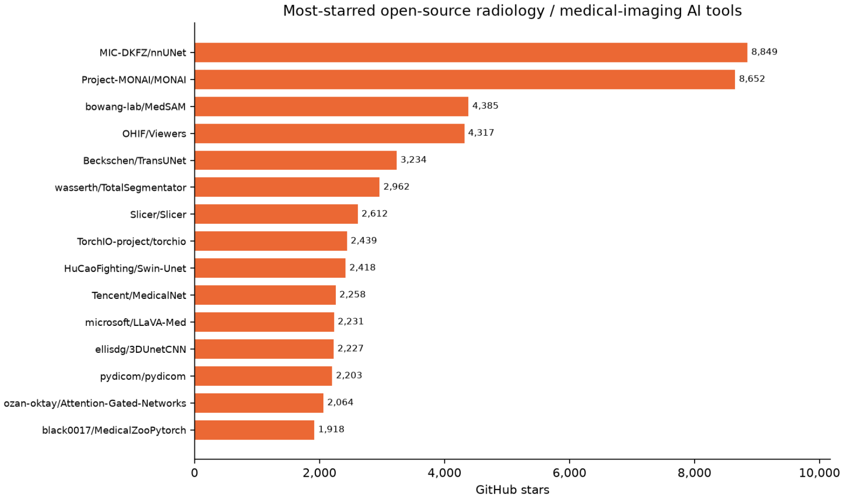
Biggest players — most-cited pediatric radiology AI papers



What the most-cited pediatric papers answer ($\geq$ 100 citations)

Cites	Paper	Clinical question it answers
472	The RSNA Pediatric Bone Age Machine Learning Challenge (2018)	*bone age*: How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
469	Performance of a Deep-Learning Neural Network Model in Assessing Skel... (2017)	*bone age*: Can a CNN assess skeletal maturity as accurately as radiologists?
454	Fully Automated Deep Learning System for Bone Age Assessment (2017)	*bone age*: Fully automated bone age from hand radiographs
434	Deep learning for automated skeletal bone age assessment in X-ray ima... (2016)	*bone age*: Deep learning for automated bone age (BoNet)
203	Improving Image Quality and Reducing Radiation Dose for Pediatric CT... (2020)	*reconstruction*: Can deep-learning reconstruction cut pediatric CT dose while improving image quality?
166	Computerized Bone Age Estimation Using Deep Learning Based Program: E... (2017)	*bone age*: Is a commercial deep-learning bone-age program accurate and faster in practice?
133	Regression Convolutional Neural Network for Automated Pediatric Bone... (2018)	*bone age*: Regression CNN for pediatric bone age
126	Artificial intelligence-assisted interpretation of bone age radiograp... (2018)	*bone age*: Does AI assistance make radiologists' bone-age reads more accurate and consistent?
116	Deep Learning Based Attenuation Correction of PET/MRI in Pediatric Br... (2019)	*reconstruction*: Can deep learning replace CT for PET/MRI attenuation correction in children?
107	Using a Dual-Input Convolutional Neural Network for Automated Detecti... (2019)	*fracture*: Can a CNN detect supracondylar fractures on pediatric elbow radiographs?

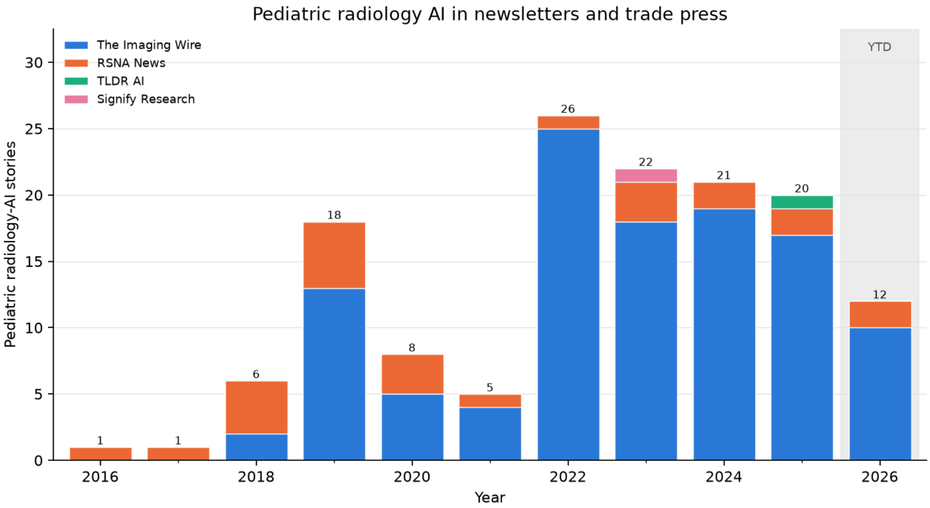
Biggest players — most-starred open-source tools



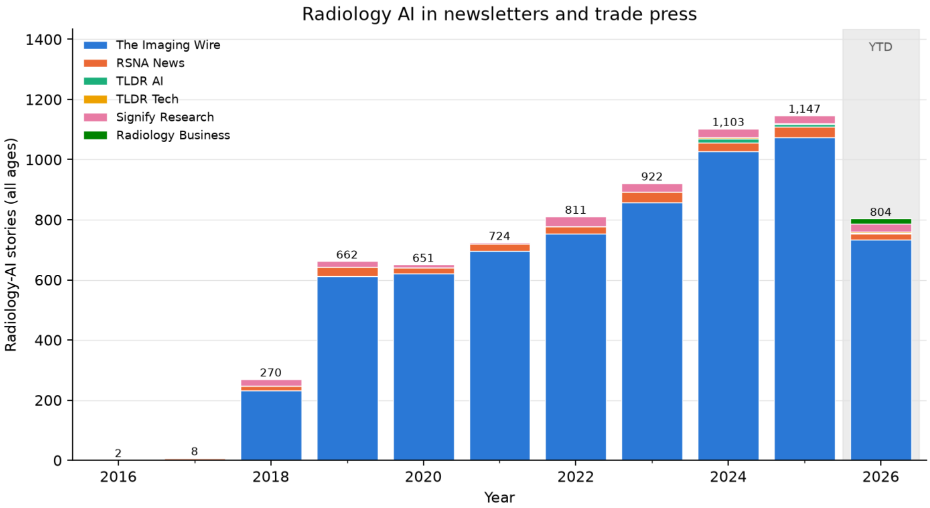
What the most-starred tools do ($\geq$ 2,000 stars)

Stars	Repository	What it is	Why it matters
8,849	MIC-DKFZ/nnUNet	self-configuring segmentation	the default strong baseline; wins most segmentation challenges out of the box
8,652	Project-MONAI/MONAI	PyTorch framework for medical imaging	the common platform most academic and vendor research models are built on (NVIDIA-backed)
4,385	bowang-lab/MedSAM	'segment anything' for medical images	click-to-segment foundation model; what interactive annotation tools are moving to
4,317	OHIF/Viewers	web DICOM viewer	the open viewer behind many AI-result overlays and research PACS
3,234	Beckschen/TransUNet	transformer U-Net	architecture paper code
2,962	wasserth/TotalSegmentator	whole-body CT / MR segmentation	free organ and bone segmentation on any CT; enables opportunistic measurements
2,612	Slicer/Slicer	3D Slicer	the free workstation for segmentation, registration, and trying AI models
2,439	TorchIO-project/torchio	3D image loading and augmentation	plumbing library
2,418	HuCaoFighting/Swin-Unet	transformer U-Net	architecture paper code
2,258	Tencent/MedicalNet	pretrained 3D backbones	transfer learning for 3D scans

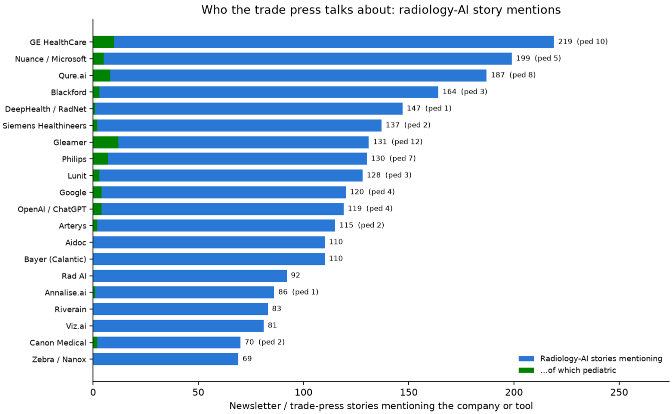
Newsletters — pediatric radiology AI stories



Newsletters — radiology AI stories, all ages

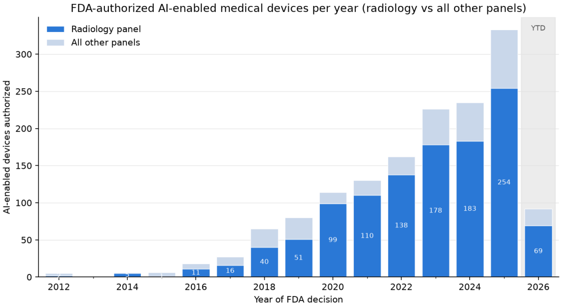


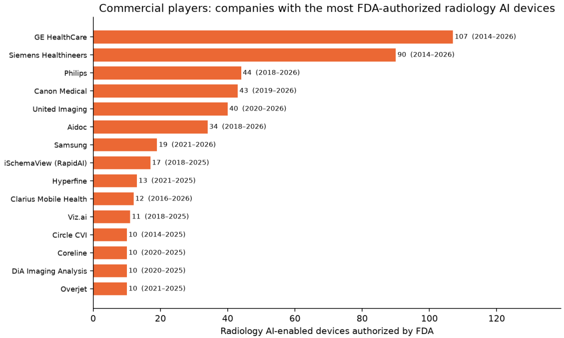
Newsletters — who the trade press talks about



A third importance signal next to citations and stars: stories mentioning each company or tool (sponsor blocks excluded). 140 pediatric radiology-AI stories across 6 archives; in 2025, 20 of 1,147 radiology-AI stories were pediatric. Leading pediatric topics: appendicitis; fetal; funding.

Commercial players — the FDA AI-enabled device list





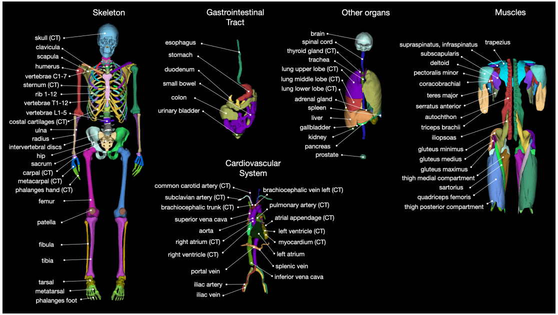
1,164 of 1,524 AI-enabled devices (76.4%) sit in the Radiology panel; 254 radiology devices were authorized in 2025. The list has no pediatric flag: age indications live in each 510(k) summary. Snapshot 2026-09-03.

Commercial software with a pediatric angle

Vendor	Product	Task	Pediatric status	FDA list
Gleamer	BoneView	fracture detection on radiographs	pediatric indication (age 2+) cleared 2023	4 (2022–2025)
AZmed	Rayvolve	fracture (and chest) findings on radiographs	pediatric fracture indication cleared 2024	4 (2022–2025)
Visiana	BoneXpert	automated bone age (GP / TW)	pediatric by design; CE-marked since 2009 (not on the FDA AI list)	not listed
16 Bit	Rho	automated bone age	pediatric by design	1 (2024)
Ever Fortune.AI	EFAI Bonesuite Bone Age Pro	automated bone age	pediatric by design	11 (2022–2026)
BrightHeart	Fetal EchoScan	fetal echocardiography view / anomaly assist	prenatal (fetal) by design	5 (2024–2025)
GE HealthCare / Canon / Siemens / Philips	TrueFidelity, AiCE, Deep Resolve, Precise Image	deep-learning CT / MR reconstruction (lower dose, faster scans)	adult-cleared scanner software, widely used for pediatric dose reduction	307 (2014–2026)
Subtle Medical / AIRS Medical	SubtleMR, SwiftMR	accelerated MRI via deep-learning denoising	adult-cleared; pediatric scan-time studies	13 (2018–2026)
Qure.ai	qXR / qER	chest radiograph and head CT triage / quantification	adult indications; pediatric TB-screening evidence only	9 (2020–2026)
Aidoc / Viz.ai	BriefCase, Viz LVO / ICH	worklist triage (hemorrhage, LVO, PE, pneumothorax)	adult-only indications; the largest deployed category	43 (2018–2026)

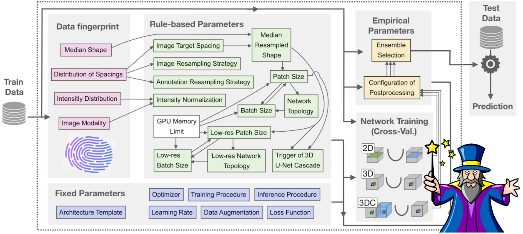
"FDA list" = number of radiology AI-enabled devices the company has on the FDA list (years). Pediatric status is the publicly stated indication; confirm the age range in the 510(k) summary before purchase.

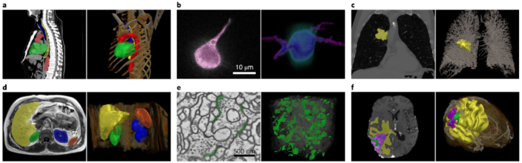
What it looks like — open-source tools



TotalSegmentator: 100+ anatomic structures segmented on any CT (repository README)

source: raw.githubusercontent.com

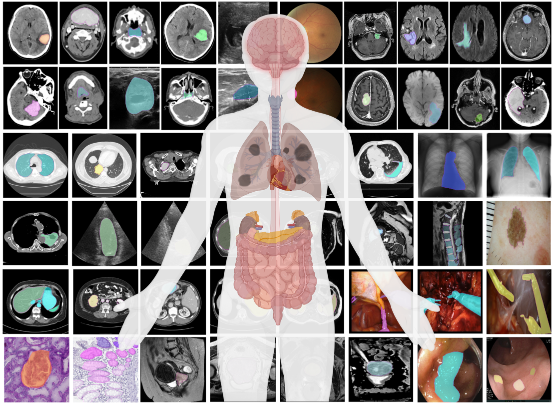




nnU-Net: self-configuring segmentation pipeline (repository README)

source: raw.githubusercontent.com

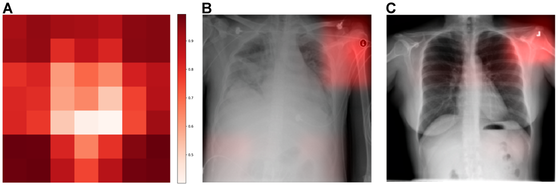
What it looks like — open-source tools



MedSAM (Nat Commun 2024, CC BY): prompt-based segmentation across modalities

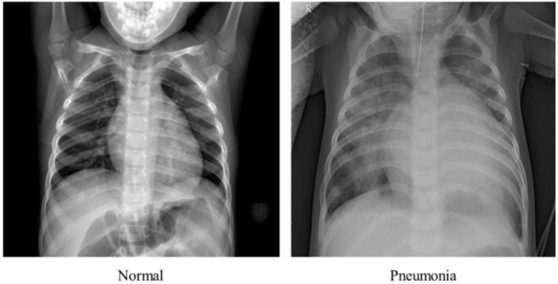
source: media.springernature.com

What it looks like — landmark papers



Zech et al. (PLoS Med 2018, CC BY): the model learns which hospital took the film

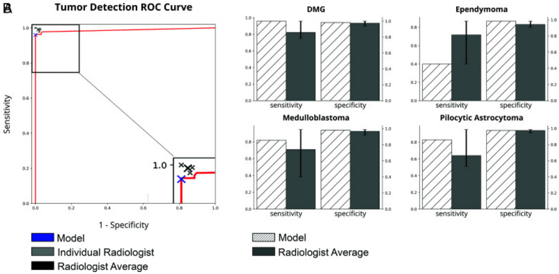
source: journals.plos.org



Pediatric chest X-ray pneumonia detection (BJR 2021), figure 1

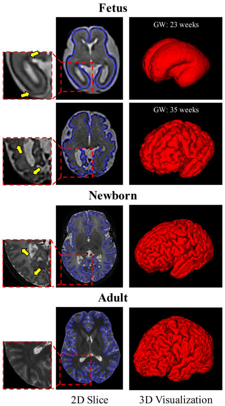
source: cdn.ncbi.nlm.nih.gov

What it looks like — landmark papers



Pediatric posterior fossa tumor detection on MRI (AJNR 2020)

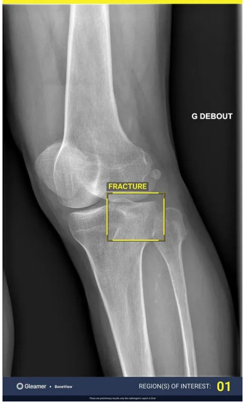
source: cdn.ncbi.nlm.nih.gov



Fetal cortical plate segmentation on MRI (IEEE TMI 2021)

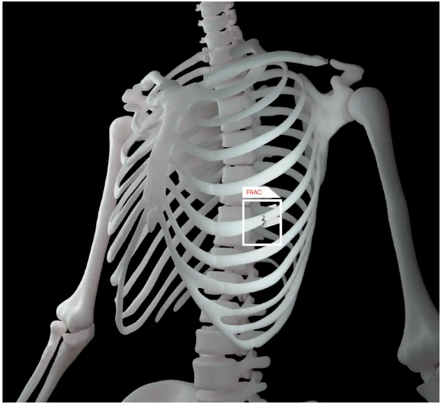
source: cdn.ncbi.nlm.nih.gov

What it looks like — commercial products



Gleamer BoneView: fracture detection with a pediatric indication (vendor site)

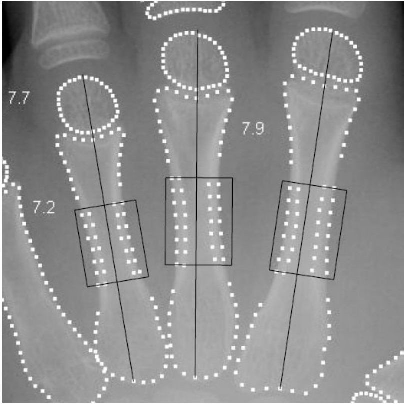
source: cdn.prod.website-files.com



AZmed Rayvolve: fracture detection on radiographs (vendor site)

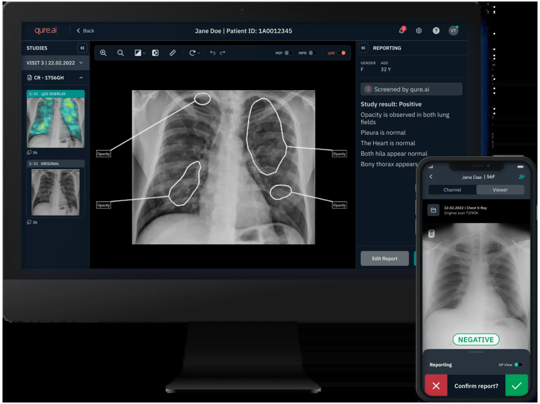
source: cdn.prod.website-files.com

What it looks like — commercial products



Visiana BoneXpert: automated bone age (vendor site)

source: bonexpert.com



Qure.ai qXR: chest radiograph findings (vendor site)

source: qure-website-images.s3.ap-south-1.amazonaws.com

Worth knowing as a radiologist — and why

- **TotalSegmentator** (open tool): free, one-command segmentation of 100+ structures on any CT (and MR); the basis for opportunistic measurements (bone density, muscle, organ volumes)
- **nnU-Net** (open tool): if a vendor claims a segmentation result, this is the baseline they had to beat
- **MONAI** (open framework): what your data scientists will build with; knowing the name lets you read a methods section
- **RSNA Pediatric Bone Age Challenge (2017)** (paper / dataset): the one pediatric task with a public benchmark, many models, and cleared products (BoneXpert, Rho)
- **Zech et al. 2018 (PLoS Med)** (paper): the cleanest demonstration that a model can learn the hospital instead of the disease; the reason to demand local validation
- **CLAIM checklist (Radiology: AI 2020)** (guideline): how to tell a well-reported AI study from a weak one
- **Brady et al. 2021 (Radiology)** (paper): deep-learning CT reconstruction lowers pediatric dose with better image quality; the most immediately usable pediatric benefit
- **FDA AI-enabled device list** (registry): the authoritative list of what is cleared; check the indication's age range before buying
- **Gleamer BoneView / AZmed Rayvolve** (products): the first fracture-detection tools with pediatric indications; realistic first pediatric AI purchases
- **MedSAM / segment-anything models** (foundation models): where annotation and segmentation tooling is heading: click, don't draw

What radiology AI does well

- **Worklist triage**: hemorrhage, large-vessel occlusion, pulmonary embolism, pneumothorax — the most validated, most deployed category (and the largest block of FDA clearances).
- **Detection / measurement aids** on high-volume adult exams (lung nodules, mammography, fractures, volumetry).
- **Image quality / acquisition**: deep-learning reconstruction and denoising — lower dose, shorter scans (most valuable in pediatrics).
- **Quantification / standardization**: segmentation, longitudinal tumor measurement, opportunistic screening from existing CTs.

Bleeding edge

- Foundation models and vision-language / report-generation systems.
- Agents: LLMs that call imaging tools and take multi-step actions (still $<$1% of papers).
- Opportunistic screening (bone density, coronary calcium, body composition).
- Multimodal and longitudinal models (imaging + EHR + priors).
- Self-supervised / label-efficient learning (key where labels are scarce).
- **Pediatric frontiers**: bone age (mature), fetal/neonatal brain MRI, congenital anomaly detection, scoliosis / Cobb-angle, growth-aware models.

What remains unresolved

- **Generalization**: models degrade across scanners, sites, populations.
- **Pediatric data scarcity & age dependence**: children are not small adults; adult models transfer poorly.
- **Prospective benefit**: most evidence is retrospective accuracy, not improved outcomes.
- **Report-generation trust**: LLM reports can be fluent and wrong.
- **Regulation/liability**: most cleared devices are validated on adults; only a handful of radiology clearances carry a pediatric indication.
- **Workflow integration, monitoring, drift, and equity.**

1. **Buy maturity, build for the gaps**: adopt cleared adult-derived tools that transfer (triage, reconstruction); treat pediatric tasks as local validation / research.
2. **Demand local pediatric validation** before clinical use, and check the age range in the clearance.
3. **Prioritize dose and throughput**: deep-learning reconstruction gives the clearest pediatric benefit today.
4. **Plan for monitoring**: pediatric drift (growth, protocol change) is faster than in adults.

Summary

- Radiology AI grew from **0.8%** to **11.4%** of the radiology literature (2008–2025), $\approx$22%/yr; 20,831 papers in 2025.
- Pediatric work is **11.3%** of radiology AI — small but growing; MRI-heavy, bone age and neuro lead.
- RSNA journals: AI share **1.1%** $\rightarrow$ **24.5%**.
- Commercial: 1,164 FDA-authorized radiology AI devices; scanner makers and triage vendors dominate; pediatric indications are rare (fracture, bone age, fetal echo).
- Mature where data are large and adult (triage, reconstruction); pediatric bone age is the one mature pediatric task.
- Generalization, pediatric data, and prospective benefit remain the gaps.